Agenda

1. Spark Core Components
2. Spark Execution Steps
3. Spark RDD
4. RDD operation
5. Demo of Spark

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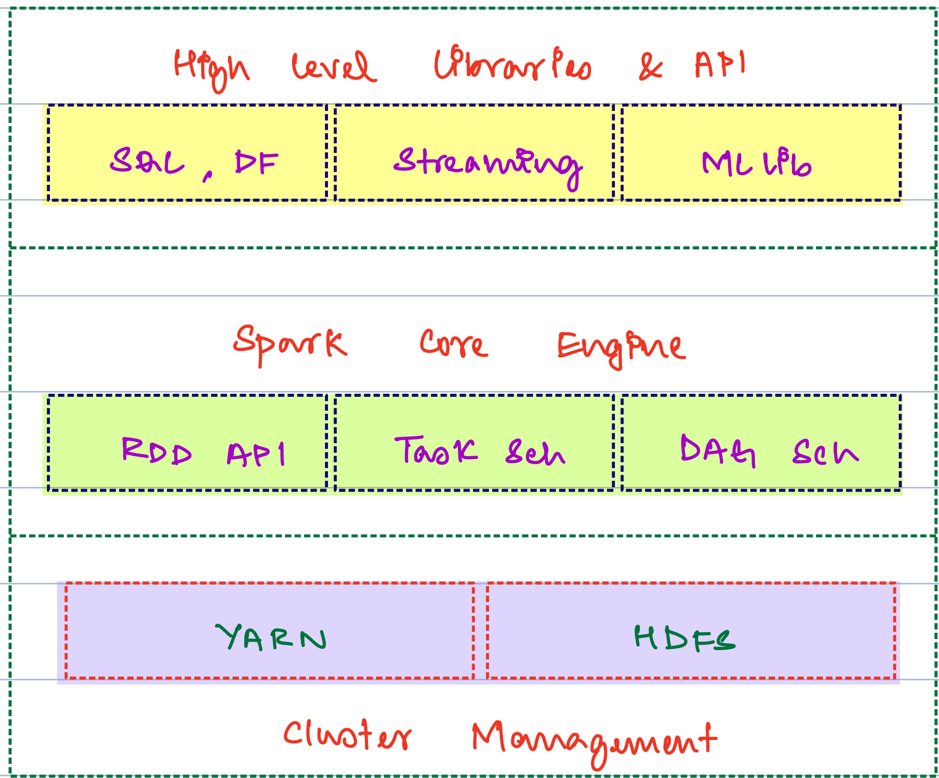
Usecase

Which comedy movie has got most ratings?

Spark components



RDD

Cluster Management

Responsible for allocating resources, provides data

Spark Core

Heart of spark. Helps us with low level abstraction

High level Libraries

Human interactable abstractions built on top of your spark core

Spark RDD → Resilient Distributed Dataset

1. Fundamental Data abstraction in Spark.
2. An Immutabe, partitioned collection of records that can be operated on parallel

once created you cannot change the value.

How to create RDD ?

RDD

- Read from File
- Parallelized Collection
- Another RDD

Read from File

usind `sc. texFile ( " path " ) ;`

Parallelized Collection

using `sc. parallelize ( List ) ;`

Another RDD

using transformation

Features of RDD

① Lazy evaluation → path does not exit

The execution of code will only happen if action is triggered

toDebug String() → park for time being

② In memory computation

Spark behaviour of processing data in RAM for a single action, discarding from memory when job is done

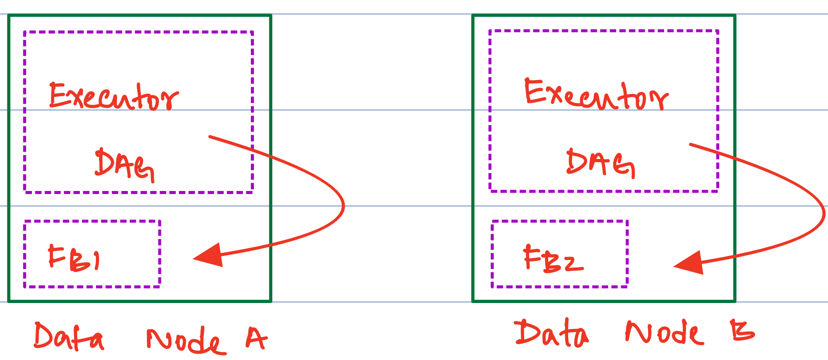
③ Fault Tolerant

If an executor goes down, spark passes the DAG to a new executor to take its place.

10:05 pm → 10:12 pm break

Visualize How RDD is created

movies.dat → 256 MB



when an action is triggered the DAG is passed to executor in Data Node.

Spark RDD operation

① Action: Forces RDD to load the data (1 Job per action)

Operation	What it returns	Purpose
take	output of limit rows	Shows limited Rows
collect	outputs all rows	Shows all rows
first	outputs first rows	Shows 1 row
count	outputs count of rows	Shows int count of rows

② Transformation : Modify RDD by performing functions

Operation	What it returns	Purpose
map	A new RDD	to apply function to input
flatmap		
filter	A new RDD	applies a where clause
group By Key	A new RDD	groups by key
join ()		

2 stages group By Key   1 stage map

Narrow Transformation

→ is self sufficient and does not need a new stage creation

→ A single partition can individually perform the function / transformation

Wide Transformation

→ is not self sufficient. We need to create a new stage while shuffling the data

→ Data will be shuffled between partitions

partition 1	partition 2
(1,1) (1,2)	(1,8) (2,6)
(2,1) (2,2)	(5,7)
(3,3)	

map () → add 5 to value

groupbykey ()

Next class

1. Tasks
2. Solve for usecase of movies Dataset